

Features of the Internet

- ⊕ It is a **digital** transmission system
 - ➡ Information is converted to symbols (zeros and ones)
 - ➡ Transmission system is designed to convey symbols
- ⊕ It is a global **network of networks**
 - ➡ WAN (Wide Area Network)
 - ➡ LAN (Local Area Network)
- ⊕ It is a **packet-switching** network
 - ➡ Transfer mode: packet switching
- ⊕ It has a **layered** network architecture
 - ➡ OSI reference model
 - ➡ TCP/IP architecture

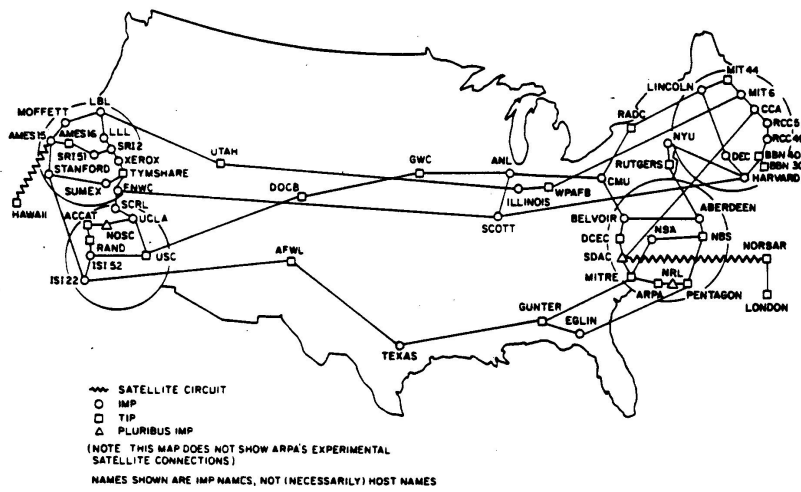
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ARPANET Advanced Research Project Agency Network (DoD)

- ⊕ ARPANET was developed to provide a test bed for researching packet-switching networks
 - ➡ Developed in the late 1960s; it was the first major effort to interconnect computers across a Wide Area Network (**WAN**)
 - ➡ For Packet-Switching Research:
 - Packet = Header + Data → control info
 - ARPANET = Packet Switches + 56 Kbps Leased Lines
 - Distributed Routing
 - Congestion Control }
 - Flow Control }
 - ➡ ARPANET led to many innovations:
 - The TCP/IP protocols as the basis for Internet
 - Several lasting applications such as Email, remote login, file transfer

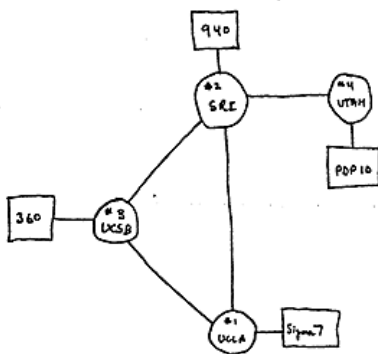
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What did ARPANET look like?



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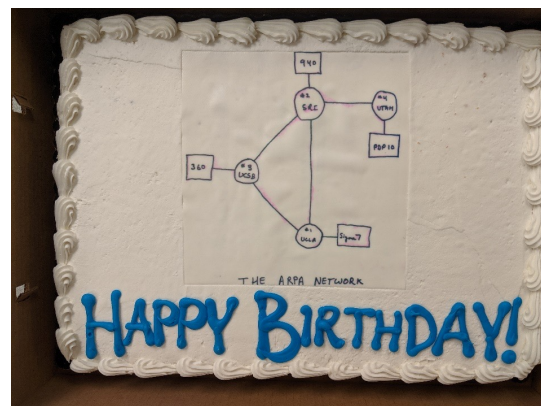
What did ARPANET look like?



THE ARPA NETWORK

DEC 1969

4 NODES



ARPANET Birthday: Oct. 29, 1969

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Local Area Network (LAN)

- ⊕ In 1980s, affordable computers become available
- ⊕ Need for low-cost, high-speed networks
 - ➡ to interconnect **local** computers
 - ➡ to access **locally** shared resources (e.g., printers, storage, servers)
- ⊕ Networks with limited distances (< 1 km) are called LANs
 - ➡ Short distance → high-speed communication with low error rate over cheap coaxial cable becomes possible
 - ➡ Messages are broadcasted to all machines in the LAN
 - Network Interface Card (NIC) of each machine has a globally unique address
 - A Medium Access Control (MAC) protocol becomes essential to coordinate access to the transmission medium

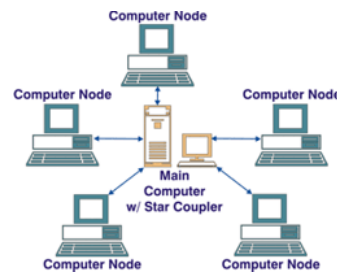
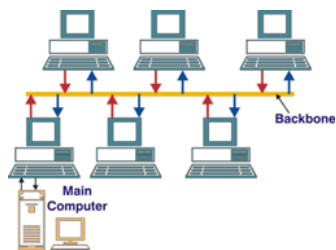
on the shared medium

*MAC Address
(48 bits)*

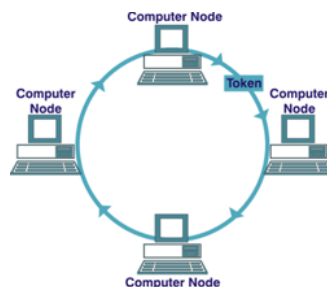
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Various LAN Topologies

*IEEE 802.3
Ethernet
Bus*



Star



*IEEE 802.5
Token Ring
Ring*

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The OSI Reference Model

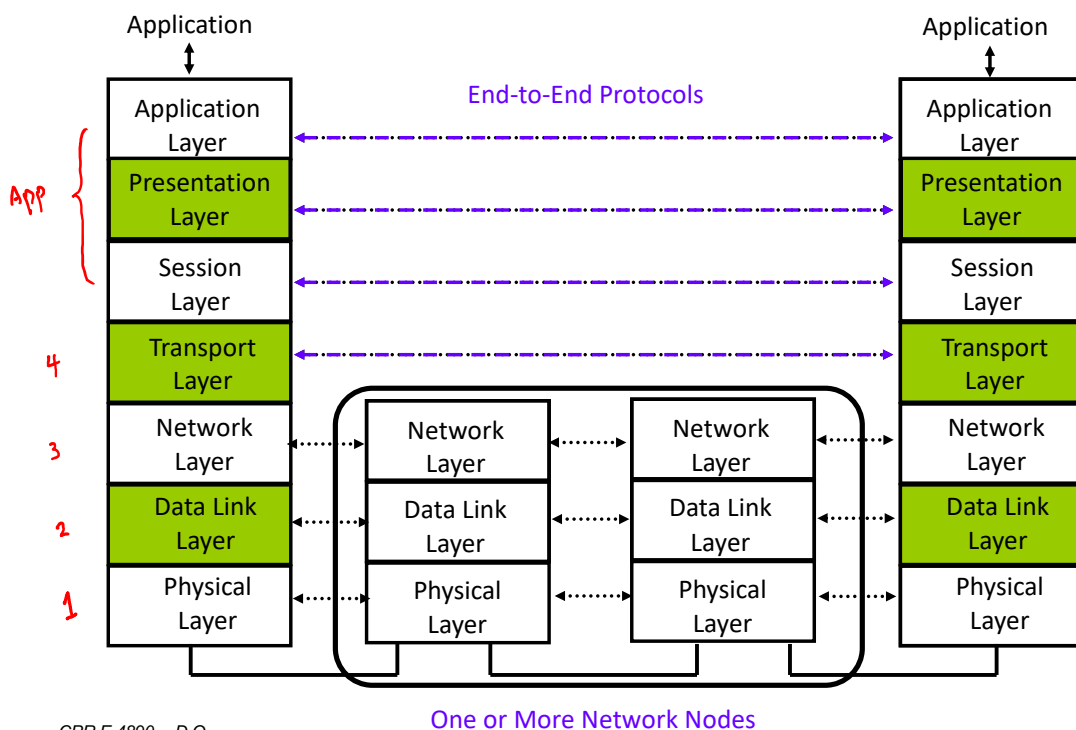
1984

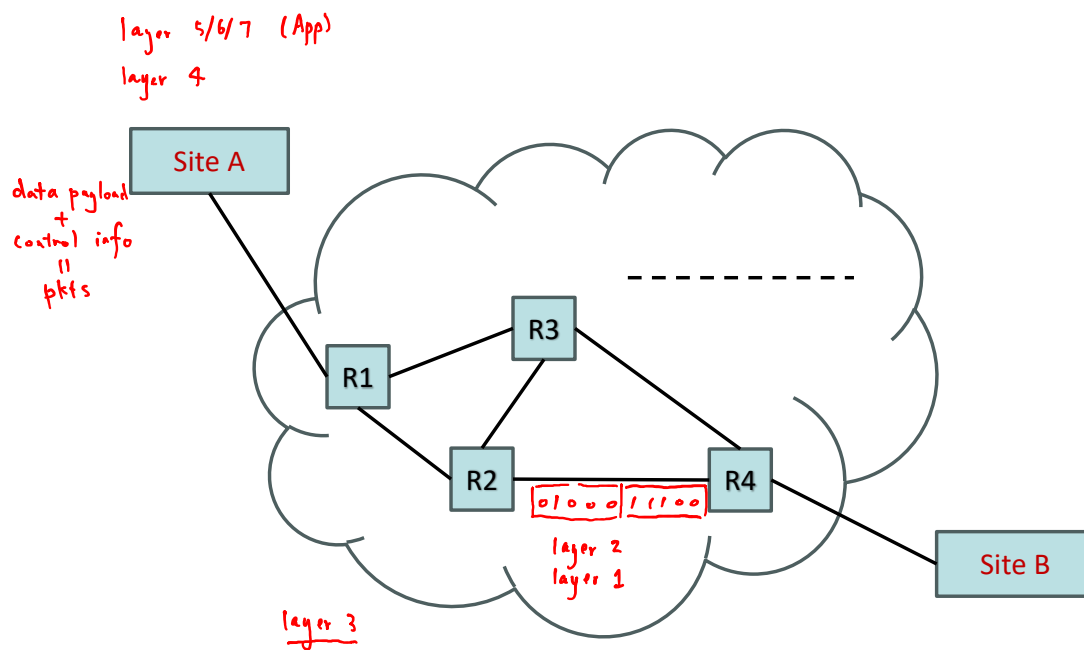
⊕ OSI (Open System Interconnection) Model

- developed by ISO (International Organization of Standardization)
- describes a **7-layer** abstract reference model for a network architecture
- provides a common framework for the development of standard protocols

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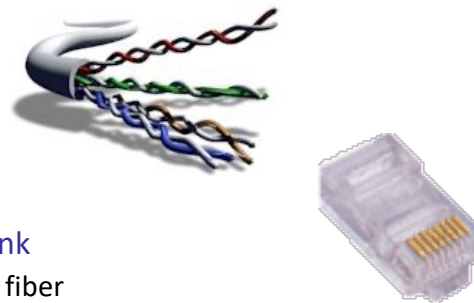
7-Layer OSI Reference Model





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Physical Layer (Layer 1)

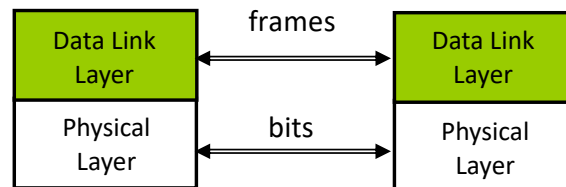


- ⊕ Transfers **bits** across a communication link
 - Twisted-pair cable, coaxial cable, optical fiber
 - Radio, infrared, ...
- ⊕ Definition and specification of the physical aspects of a communication link
 - Mechanical: cable, plugs, pins...
 - Electrical/optical: modulation, signal strength, voltage levels, ... }
 - Functional/procedural: how to activate, maintain, and deactivate physical links, ... }

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Data Link Layer (Layer 2)

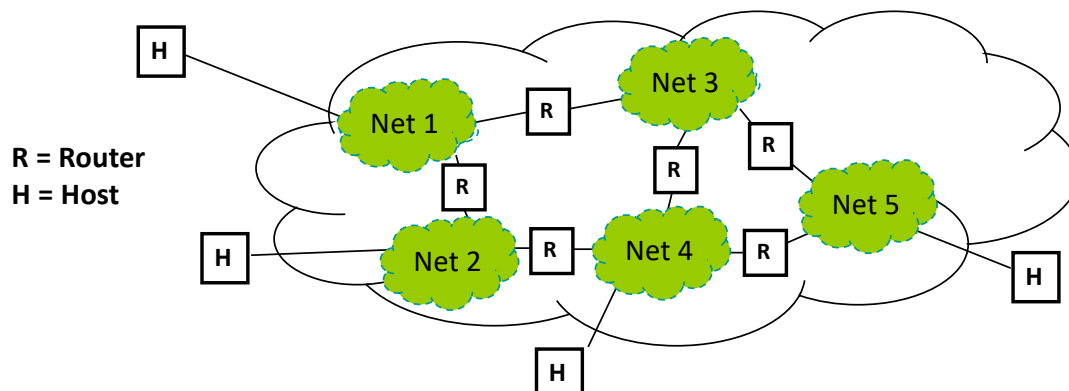
- ⊕ Transfers **frames** ^{blocks of bits} across direct connections ^{communication link}
- ⊕ Framing: groups bits into frames
- * ⊕ Detection of bit errors; retransmission of frames
- ⊕ Flow Control ✓ ^{no congestion control}
- ⊕ Medium Access Control for LANs (Local Area Networks)
^{MAC Protocol}



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Network Layer (Layer 3)

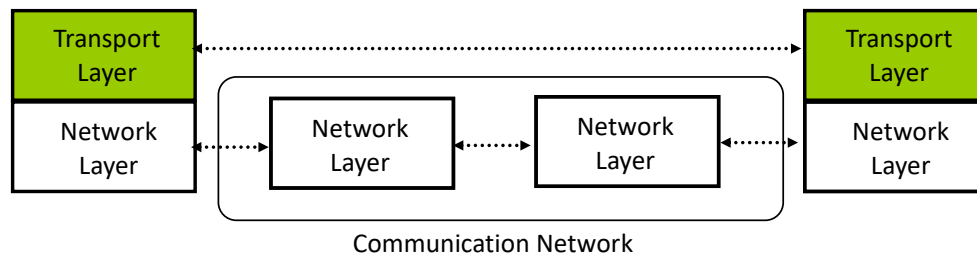
- ⊕ Transfers **packets** across multiple links and/or multiple networks
- ⊕ Addressing must scale to large networks
- ⊕ Nodes jointly execute **routing** algorithm to determine paths across the network
- ⊕ Best-effort connectionless service (no guarantee)



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Transport Layer (Layer 4)

- Transfers **segments** from process in one machine to process in another machine (end-to-end transfer)
 - * Reliable stream transfer or quick-and-simple single-block transfer
 - Port numbers enable multiplexing
 - Connection setup, maintenance, and release
 - Congestion control, flow control
- Handwritten notes:*
TCP (next to first two items)
UDP (next to 'single-block transfer')
TCP/21: FTP
TCP/80: HTTP
Handshake (with arrow pointing to 'Connection setup, maintenance, and release')



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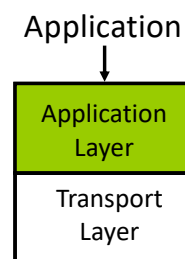
Upper Layers (Layers 5, 6, 7)

- Application Layer: provides services that are frequently required by applications – DNS, web access, file transfer, email, ...

- ~~Presentation Layer: machine-independent representation of data, ...~~

- ~~Session Layer: dialog management, recovery from errors, ...~~

Been incorporated into
Application Layer

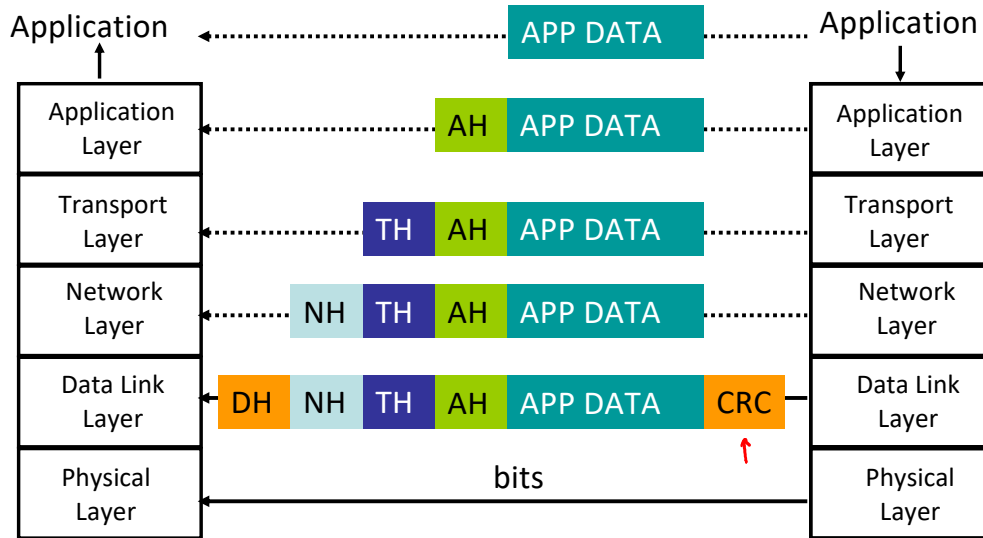


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Headers & Trailers

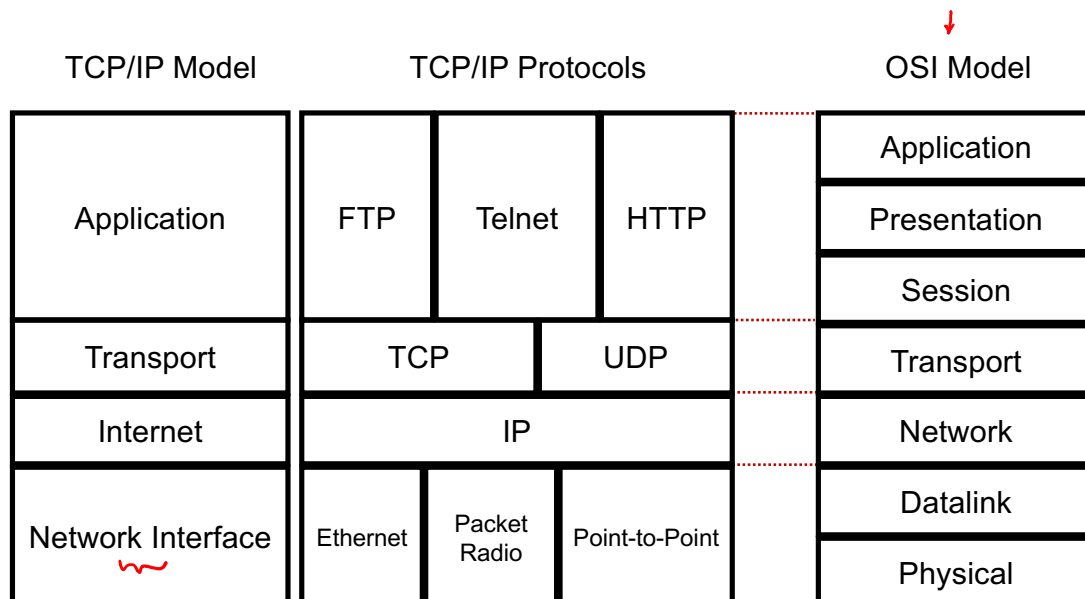
Encapsulation, overhead in headers and trailer

- Each protocol uses a header to carry control information such as addresses, sequence numbers, flag bits, length indicators, etc.
- CRC check bits are **appended** at Data Link Layer for error detection



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TCP/IP Network Architecture



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